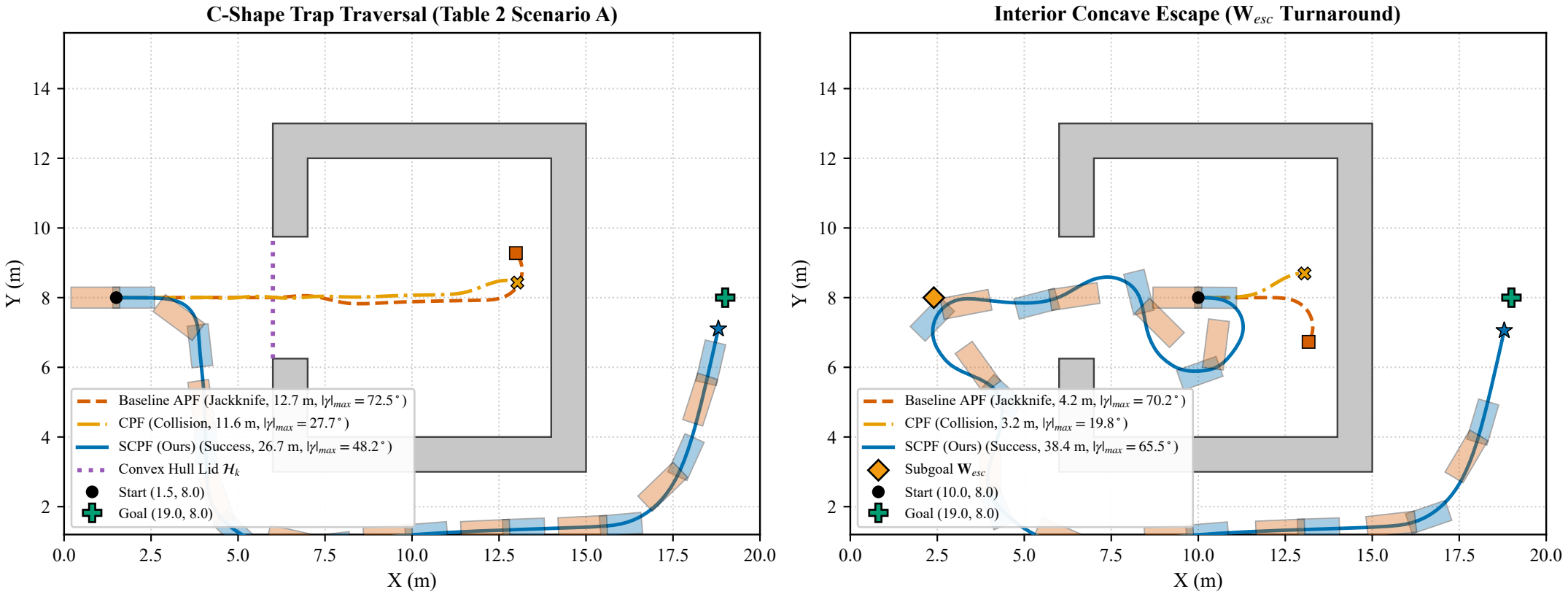

C-Shape Trap Traversal (Table 2 Scenario A)
Interior Concave Escape (W_{esc} Turnaround)
Y (m)
X (m)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
2
4
6
8
10
12
14
Baseline APF (Jackknife, 12.7 m, $|\gamma|_{max} = 72.5^\circ$)
CPF (Collision, 11.6 m, $|\gamma|_{max} = 27.7^\circ$)
SCPF (Ours) (Success, 26.7 m, $|\gamma|_{max} = 48.2^\circ$)
Convex Hull Lid $\mathcal{H}_k$
Start (1.5, 8.0)
Goal (19.0, 8.0)
Baseline APF (Jackknife, 4.2 m, $|\gamma|_{max} = 70.2^\circ$)
CPF (Collision, 3.2 m, $|\gamma|_{max} = 19.8^\circ$)
SCPF (Ours) (Success, 38.4 m, $|\gamma|_{max} = 65.5^\circ$)
Subgoal $\mathbf{W}_{esc}$
Start (10.0, 8.0)
Goal (19.0, 8.0)